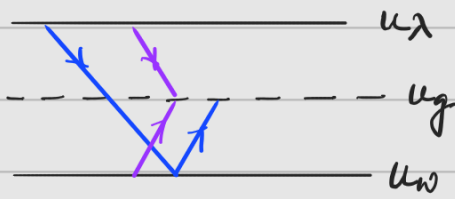


1)



$$\sigma = \frac{\tau_i - \tau_r}{\tau_i - \tau_w}$$

$$u_g = \frac{1}{2} u_\lambda + \frac{1}{2} u_{refl}$$

$$u_{refl} = (1 - \sigma) u_\lambda + \sigma u_w$$

$$\begin{aligned} \therefore u_g &= \frac{1}{2} u_\lambda + \frac{1}{2} [(1 - \sigma) u_\lambda + \sigma u_w] \\ &= \frac{1}{2} (2 - \sigma) u_\lambda + \frac{\sigma}{2} u_w \end{aligned}$$

Taylor Series expansion of velocity profile near wall:

$$\sigma(u_\lambda - u_w) = \left. \frac{du}{dy} \right|_w \cdot 2\lambda + \dots$$

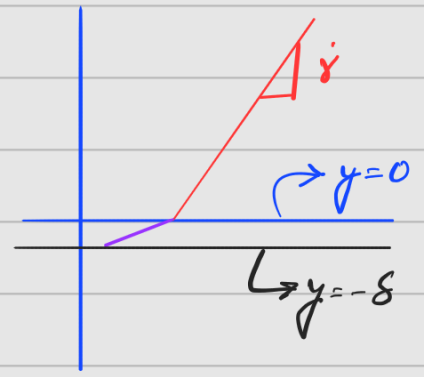
$$\Rightarrow u_\lambda = u_w + \frac{2\lambda}{\sigma} \left. \frac{\partial u}{\partial y} \right|_w$$

$$\therefore u_g = \left[\frac{u_w}{2} + \frac{\lambda}{\sigma} \left. \frac{\partial u}{\partial y} \right|_w \right] (2 - \sigma) + \frac{\sigma}{2} u_w$$

$$\Rightarrow u_g = u_w + \frac{2 - \sigma}{\sigma} \lambda \left. \frac{\partial u}{\partial y} \right|_w$$

$$\text{For this case, } u_g = \frac{2 - \sigma}{\sigma} \lambda \left. \frac{\partial u}{\partial y} \right|_w$$

GDE: $0 = -\cancel{\frac{dp}{dx}} + \mu \frac{d^2 u}{dy^2}$



$$\therefore u_x = C_2 y + C_3 \Rightarrow \frac{du_x}{dy} = C_2$$

$$u_y = C_3 y + C_4 \Rightarrow \frac{du_y}{dy} = C_3$$

BC:

1. From imposed Shear rate for the liquid layer:

$$\frac{du_x}{dy} = \dot{\gamma} \Rightarrow C_2 = \dot{\gamma}$$

2. Stress continuity at the interface:

$$\mu_x \frac{du_x}{dy} = \mu_y \frac{du_y}{dy} \Rightarrow \mu_x C_2 = \mu_y C_3$$

$$\Rightarrow C_3 = \frac{\mu_x \dot{\gamma}}{\mu_y}$$

3. Maxwell 1st order slip at bottom wall:

$$u_y(-\delta) = \frac{2-\sigma}{\sigma} \lambda \left. \frac{du_y}{dy} \right|_{y=-\delta} \Rightarrow -C_3 \delta + C_4 = \frac{2-\sigma}{\sigma} \lambda C_3$$

$$\Rightarrow C_4 = \left(\frac{2-\sigma}{\sigma} \lambda + \delta \right) \frac{\mu_x \dot{\gamma}}{\mu_y}$$

4. Velocity continuity at the interface:

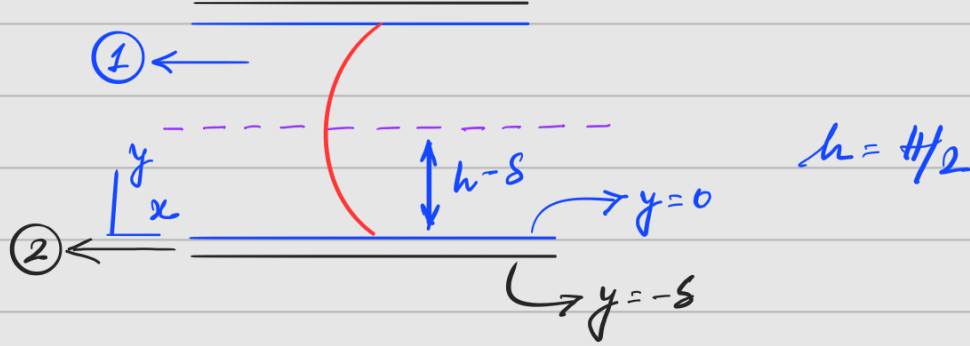
$$u_y(0) = u_x(0) = 0 \Rightarrow C_2 = C_4 = \left(\frac{2-\sigma}{\sigma} \lambda + \delta \right) \frac{\mu_x}{\mu_y} \dot{\gamma}$$

$$\therefore u_x = \dot{\gamma} y + \left(\frac{2-\sigma}{\sigma} \lambda + \delta \right) \frac{\mu_x}{\mu_y} \dot{\gamma}$$

For slip length l_s , $u_x(-l_s) = 0$

$$\therefore l_s = \frac{\mu_x}{\mu_y} \left(\frac{2-\sigma}{\sigma} \lambda + \delta \right)$$

2)



$$\text{GDE : } \mu \frac{d^2 u}{dy^2} - \frac{dp}{dx} = 0$$

$$\Rightarrow \frac{d^2 u}{dy^2} = \frac{1}{\mu} \frac{dp}{dx}$$

$$\Rightarrow u_1 = \frac{1}{\mu_1} \frac{dp}{dx} \frac{y^2}{2} + C_1 y + C_2$$

$$u_2 = \frac{1}{\mu_2} \frac{dp}{dx} \frac{y^2}{2} + C_3 y + C_4$$

BC :

$$1. \text{ Symmetry at } y = h-s \Rightarrow \frac{du_1}{dy} = 0$$

$$\Rightarrow \frac{1}{\mu_1} \frac{dp}{dx} (h-s) + C_1$$

$$\Rightarrow C_1 = -\frac{1}{\mu_1} \frac{dp}{dx} (h-s)$$

2. Continuity of shear stress at interface :

$$\Rightarrow \mu_1 \frac{du_1}{dy} = \mu_2 \frac{du_2}{dy} \Rightarrow \mu_1 C_1 = \mu_2 C_3 \Rightarrow C_3 = \frac{\mu_1}{\mu_2} C_1$$

$$\therefore c_3 = -\frac{1}{\mu_2} \frac{dp}{dx} (h-\delta)$$

3. No slip at wall:

$$\frac{1}{\mu_2} \frac{dp}{dx} (-\delta)^2 + c_3(-\delta) + c_4 = 0$$

$$\Rightarrow c_4 = -\frac{1}{\mu_2} \frac{dp}{dx} (h-\delta)\delta - \frac{1}{\mu_2} \frac{dp}{dx} \frac{\delta^2}{2}$$

$$= -\frac{1}{\mu_2} \frac{dp}{dx} \left(h\delta - \frac{\delta^2}{2} \right)$$

4. Velocity continuity:

$$u_1|_{y=0} = u_2|_{y=0}$$

$$\Rightarrow c_2 = c_4 = -\frac{1}{\mu_2} \frac{dp}{dx} \left(h\delta - \frac{\delta^2}{2} \right)$$

$$\therefore u_1 = \frac{1}{\mu_1} \frac{dp}{dx} \frac{y^2}{2} + c_1 y + c_2$$

$$= \frac{1}{\mu_1} \frac{dp}{dx} \frac{y^2}{2} - \frac{1}{\mu_1} \frac{dp}{dx} (h-\delta)y - \frac{1}{\mu_2} \frac{dp}{dx} \left(h\delta - \frac{\delta^2}{2} \right)$$

$$u_{av}(h-\delta) = \int_0^{h-\delta} u_1 dy = \frac{1}{\mu_1} \frac{dp}{dx} \frac{(h-\delta)^3}{2 \times 3} - \frac{1}{\mu_1} \frac{dp}{dx} \frac{(h-\delta)^3}{2} - \frac{1}{\mu_2} \frac{dp}{dx} \left(h\delta - \frac{\delta^2}{2} \right) (h-\delta)$$

$$\therefore u_{av} = \frac{1}{\mu_1} \frac{dp}{dx} \frac{(h-s)^2}{6} - \frac{1}{\mu_1} \frac{dp}{dx} \frac{(h-s)^2}{2} \\ - \frac{1}{\mu_2} \frac{dp}{dx} \left(hs - \frac{s^2}{2} \right)$$

$$= \frac{1}{\mu_1} \left(-\frac{dp}{dx} \right) \frac{(h-s)^2}{3} + \frac{1}{\mu_2} \left(-\frac{dp}{dx} \right) \left(hs - \frac{s^2}{2} \right)$$

$$\approx \left(-\frac{dp}{dx} \right) \frac{h}{3\mu_1} \left(h + \frac{3\mu_1 s}{\mu_2} \right)$$

$$\tau_w = \mu_1 \frac{du_1}{dy} = \mu_1 G = \mu_1 \frac{G}{\mu_1} (h-s) \approx -\frac{dp}{dx} (h-s)$$

$$\approx \frac{u_{av} (h-s)}{\frac{h}{3\mu_1} \left(h + \frac{3\mu_1 s}{\mu_2} \right)} \approx \frac{u_{av} 3\mu_1}{h + \frac{3\mu_1 s}{\mu_2}}$$

$$F_{visc} = \tau_w 2ws \approx \frac{6w\mu_1 s}{h + \frac{3\mu_1 s}{\mu_2}} u_{av}$$

$$F_{st} = 2w\sigma \cos\theta$$

$$\frac{d}{dt}(M\bar{v}) = F_{st} - F_{visc} \approx 0$$

$$\Rightarrow \cancel{2w\sigma \cos\theta} = \frac{\cancel{6w\mu_1}}{h + \frac{3\mu_1 s}{\mu_2}} \frac{s ds}{dt}$$

$$\Rightarrow \frac{s^2}{2} = \frac{\sigma \cos\theta}{3\mu_1} \left(h + \frac{3\mu_1 s}{\mu_2} \right) t$$

$$\Rightarrow S = \sqrt{\frac{2\sigma\omega\cos\theta}{3\mu_1} \left(\mu + \frac{3\mu_1 S}{\mu_2} \right) t}$$

$$= \sqrt{\frac{2\sigma h\omega\cos\theta}{3\mu_1} \left(1 + \frac{3\mu_1 S}{\mu_2 h} \right) t}$$